

R version 4.6.0 (2026-04-24 ucrt) -- "Because it was There"
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 Platform: x86_64-w64-mingw32/x64

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Natural language support but running in an English locale

R is a collaborative project with many contributors.
 Type 'contributors()' for more information and
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Type 'demo()' for some demos, 'help()' for on-line help, or
 'help.start()' for an HTML browser interface to help.
 Type 'q()' to quit R.

[Previously saved workspace restored]

```
> library(readxl)
> library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
> library(writexl)
>
> # =====
> # PATH
> # =====
>
> tone_file <- "D:/SKRIPSI/TONE.xlsx"
>
> fundamental_file <- "D:/SKRIPSI/Fundamental.xlsx"
>
> output_file <- "D:/SKRIPSI/IM_TONE.xlsx"
>
> # =====
> # BACA DATA
> # =====
>
> tone <- read_excel(tone_file)
-/
>
> fundamental <- read_excel(fundamental_file)
-/
>
> # =====
> # MERGE DATA
> # =====
>
> data_merge <- tone %>%
+   left_join(
+     fundamental,
+     by = c("Firm_Code", "Year")
+   )
>
```

```

> # =====
> # HAPUS OBSERVASI YANG TIDAK LENGKAP
> # =====
>
> reg_data <- data_merge %>%
+   filter(
+     !is.na(TONE),
+     !is.na(EARN),
+     !is.na(SIZE),
+     !is.na(BTM),
+     !is.na(RET)
+   )
>
> # =====
> # REGRESI TONE FUNDAMENTALS
> # =====
>
> tone_model <- lm(
+   TONE ~ EARN + SIZE + BTM + RET,
+   data = reg_data
+ )
>
> # =====
> # HITUNG IM_TONE (RESIDUAL)
> # =====
>
> reg_data$IM_TONE <- residuals(tone_model)
>
> # =====
> # GABUNGAN KEMBALI KE DATA AWAL
> # =====
>
> hasil_final <- data_merge %>%
+   left_join(
+     reg_data %>%
+       select(
+         Firm_Code,
+         Year,
+         IM_TONE
+       ),
+     by = c("Firm_Code", "Year")
+   )
>
> # =====
> # EXPORT EXCEL
> # =====
>
> write_xlsx(
+   hasil_final,
+   output_file
+ )
>
> # =====
> # RINGKASAN
> # =====
>
> cat("\n=====\\n")
=====
> cat("SELESAI\\n")
SELESAI
> cat("Jumlah Observasi :", nrow(hasil_final), "\\n")
Jumlah Observasi : 328
> cat("Output :", output_file, "\\n")
Output : D:/SKRIPSI/IM_TONE.xlsx
> cat("=====\\n")
=====
>

```

```
> summary(tone_model)
```